

DEID annotation guidelines (EN)

ProductionLabels_v1

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Name Patient Caregiver Other Address_Location Patient Caregiver Other Organization Healthcare Other	Date Age_Birthdate Profession Contactdetails ID Patient Caregiver Anonymize_Other

Name

This label covers all personal names and initials that refer to people. It is subdivided into Name:Patient, Name:Caregiver and Name:Other.

Name:Patient

The patient is the person the record belongs to. First names, surnames and initials must all be annotated as Name:Patient.

- Forms of address such as Mr/Mrs must not be annotated.
- Generic terms such as patient must not be annotated.

Examples:

- 1) "Jan Jansen was admitted to the ICU"
- 2) "Veerle had chest pain"
- 3) "K. Aerts was seen at the cardiology clinic"
- 4) "Mr De Meyer received chemotherapy"
- 5) "The patient felt well"
- 6) "Pt is dizzy"
- 7) "The lady preferred not to be treated"
- 8) "Jan's mother is worried"

Name:Caregiver

These are the names and initials of care providers, meaning any healthcare worker who comes into contact with the patient. This includes doctors, nurses, students and allied health professionals such as physiotherapists, speech therapists, dietitians, etc., as well as social services staff. External doctors and care providers also belong here.

- Professional titles such as dr./prof. or doctor are annotated together with the name when they appear immediately before or after it. If the title cannot be captured in a single annotation with the name because another word comes between them, do not annotate the title.
- Job descriptions referring to a specialty (e.g. pulmonologist) are not annotated.
- Roles such as head of department, responsible supervisor, ... are not annotated.
- Names of care providers that form part of a publication reference are not annotated. This choice was made because publications are publicly available, which creates a route to re-identifying a care provider or a date.

Examples:

- 1) "Validated by Dr. Geraert"
- 2) "Head of department Prof. Dr. Iepermans"
- 3) "Aaron inserted the IV line"
- 4) "Radiologist Jan Aerschot, Doctor"
- 5) "Pl worked the night shift." (Initials refer to care provider Peter Ijkert)
- 6) "Contact has already been made with Jan from Bauren Care Home" (Bauren Care Home = Organization:Healthcare)
- 7) "Jozef is a doctor" (The professional title doctor is not directly adjacent to the name because other words come between them → do not annotate the title)
- 8) "Student JK saw the patient this morning" (JK = Name:Caregiver)
- 9) "(Friedmans et al. 2012)" (Even though Friedmans may be a doctor, do not annotate. The date is not annotated here either, because of the increased re-identification risk.)

Name:Other

These are names and initials of family members, friends, other patients whose record this is not, and external people who are not care providers. The only exception to this rule is in reproductive medicine, where a patient together with their partner and/or child can both be considered patients within the same record.

Examples:

- 1) "Sarah is the patient's mother"
- 2) "Joost had an argument with a fellow patient, Ronald" (Joost = Name:Patient)
- 3) "Lies is coming in for a NIPT test together with her husband Jacob" (Lies = Name:Patient, Jacob = Name:Patient because this is reproductive medicine)

Address Location

This label covers all descriptions of addresses, such as street names, postcodes, buildings, cities, municipalities and provinces. Post office district numbers and box (unit) numbers in apartment buildings also belong here. A further distinction is made between Address_Location:Patient, Address_Location:Caregiver and Address_Location:Other.

- Countries, continents and cross-border regions with more than 100,000 inhabitants must not be annotated, UNLESS they form part of a larger address. For the annotator's convenience, anything smaller than a country may be annotated.

Address_Location:Patient

These are addresses officially linked to the patient, for example home addresses, places of birth, etc.

Examples:

- 1) "Registered address: Veltwijcklaan 77, 2180 Ekeren"
- 2) "The patient lives at Langestraat 128"
- 3) "She is of South African origin." (Country)
- 4) "She is originally from the Kempen region" (because it is smaller than a country)
- 5) "Patient's home address: Gravenlaan 23 box 2003, B-2900 Antwerp 4, Belgium." (Belgium is a country BUT part of a larger address; '4' is the post office district number)

Address_Location:Caregiver

These are addresses of care providers, the care institution, etc.

- Departments and names of care institutions that contain a place name do not belong to this label. They fall under Organization:Healthcare instead.

Examples:

- 1) "Out-of-hours GP post in Bornhem"
- 2) "The patient is taken to Middelheim General Hospital" (Middelheim General Hospital = Organization:Healthcare)
- 3) "Private practice: WILRIJK" (Private practice: WILRIJK = Organization:Healthcare)

Address_Location:Other

These are addresses of external bodies, events (e.g. accidents, travel), friends and family.

Examples:

- 1) "The accident happened at the junction of Leerstraat and Uifellaan"
- 2) "The patient went on holiday to Morocco." (Country)
- 3) "He is from the Middle East" (because this multi-country region has more than 100,000 inhabitants)
- 4) "His uncle has a holiday home in Nice" (because it is smaller than a country)

Organization

A distinction is made between Organization:Healthcare and Organization:Other because it may later be decided to keep care institutions visible after all (with targeted pseudonymisation). It can be clinically relevant that the patient attended a university hospital or a burn centre, whereas other organisations such as companies, hobby clubs, etc. are of no clinical importance and only add re-identification risk.

Organization:Healthcare

All names of care institutions relevant to the patient, such as hospitals, care homes, out-of-hours GP posts, health insurance funds, social services, etc. Annotate the full name of the

organisation. The only exception is national health authorities, because they carry as much information as a country, which is likewise not annotated.

- Department, building and room codes such as D2 or A3 are also annotated.
- Hospital route numbers are also annotated.
- Software applications specific to a hospital are also annotated.
- File names (not URLs) that refer to a care institution also fall under this label.
- Names of medical device manufacturers are not annotated.

Examples:

- 1) "The patient was admitted to Antwerp University Hospital (UZA)." (If an Address_Location sits directly next to the Organization, annotate them as a single span)
- 2) "Consultation at UZA - Edegem"
- 3) "The patient was admitted to UZA in Antwerp." (If words come between the Organization and the Address_Location, annotate them separately; Antwerp = Address_Location:Caregiver)
- 4) "The MDT meeting at MOCA takes place on Wednesday" (MOCA = the hospital's oncology centre, traceable to Antwerp University Hospital)
- 5) "There has been consultation with the burn centre at Middelheim." (the generic term burn centre is preserved; only the hospital name makes it identifiable)
- 6) "She was referred to the Pulderbos rehabilitation centre." (the generic term rehabilitation centre is preserved)
- 7) "Contact was made with the CM health insurance fund."
- 8) "An appointment was scheduled at Leuven University Hospital." (the generic term university hospital is preserved)
- 9) "He was referred to the Merksem out-of-hours GP post." (the generic term out-of-hours GP post is preserved)
- 10) "Report forwarded to the national Ministry of Health." (not annotated, because it is a national authority)
- 11) "Reimbursement by the national health insurance institute" (not annotated, because it is a national authority)
- 12) "Vaccination status is available on the national patient portal" (not annotated, because it is a national system)
- 13) "He received support from the Ekeren municipal social welfare service." (social services)
- 14) "The patient is on ward D2 after coming from the ICU." (ward codes are annotated; the word 'ward' itself is not)
- 15) "The ward is located on route 11" (hospital route)
- 16) "Surgery in OR 4" (building code)
- 17) "The patient is in UZA I4:BOX5" (= Organization:Healthcare, because it is a room code)
- 18) "Performed by MOLIS:LABO" (software application used in particular hospitals)
- 19) "Originally from C2M CHS" (software application used in particular hospitals)
- 20) "Internal guideline Pneumonia-UZA-guideline.pdf" (this refers to the care institution)

- 21) "Internal guideline pneumonia-guideline.pdf" (No reference to a care institution OR a patient)
- 22) "A Medtronic Inceptiv™ spinal cord neurostimulation system was implanted" (do not annotate Medtronic, because it is a medical device manufacturer)

Organization:Other

All organisations not relevant to the patient's treatment and follow-up.

Examples:

- 1) "He works at **ING Bank.**"
- 2) "The patient is a member of **Riviera Korfbal Club.**"
- 3) "His daughter studies at **Ghent University.**"
- 4) "The patient is a member of the **pensioners' association.**"
- 5) "She was in contact with her employer, **BASF Antwerp.**"
- 6) "He volunteers for **Red Cross Flanders.**" (→ no clinical relevance → Other)
- 7) "He sustained a skull fracture during the fire at **De Zwaan pub** in **2019.**" (De Zwaan pub = Organization:Other, 2019 = Date; 'fire' is generic enough)

Date

This covers dates in both numeric and fully written-out form: anything representing a day, month and/or year.

- Public holidays such as Christmas also count as a date.
- Seasons and clock times are not annotated.
- Weekdays (e.g. Monday) are only annotated when they appear immediately before or after a date.
- Additional information such as 'early'/'late' [month] is also annotated (e.g. early January).
- Dates are always annotated separately, UNLESS one date carries information that is essential to the other (e.g. 23/09 – 2/10/2023 → annotate together).
- By way of exception, dates are not annotated when they refer to publications, copyright statements, clinical scores or external systems, because they stand apart from the patient and the organisation AND pose a re-identification risk, since this information is publicly available.

Examples:

- 1) "The operation took place on **Monday 12/03/2021.**" (weekday → annotate, because it sits next to a date)
- 2) "Consultation scheduled for **15 August 2020.**"
- 3) "He was seen on **1 January.**"
- 4) "The procedure took place at **Christmas.**"
- 5) "Rehabilitation started in **April 2019.**"
- 6) "Discharge took place on **New Year's Eve.**"
- 7) "He came in for a check-up on Monday." (weekday → do not annotate, because it stands alone)
- 8) "She felt worse in the winter." (season → do not annotate)

- 9) "Pt came to the emergency department on 1/03/22 at 11:55" (time → do not annotate)
- 10) "He presented in early June." ('early' adds information about the time)
- 11) "Admitted between 23/09 – 2/10" (no shared information → separate)
- 12) "Admitted between 23/09 – 2/10/2023" (shared information in the final date → together!)
- 13) "Admitted from 23/09 to 2/10/2023" (shared information in the final date → together!)
- 14) "Allergy testing during the period April-June" (a period spanning several months)

Age Birthdate

This label combines date of birth (in the same formats as Date) and age.

- For ages, the unit of time (year, months, -year-old) must be annotated along with the number, but additional words such as '... old' must not.
- Gestational age is not treated as age.

Examples:

- 1) "The patient is 23 years old."
- 2) "A 65-year-old man was admitted."
- 3) "The baby is 6 months old."
- 4) "She was born on 12/05/1998."
- 5) "Date of birth: 3 October 1975."
- 6) "This concerns an 84-year-old lady."
- 7) "A boy aged 14 years came to the clinic."
- 8) "The patient has lived in Belgium for 12 years." (a duration, but not an age)
- 9) "Term baby (39w) with Apgar 8-9-9" (gestational age / age)

Profession

The occupation or job title must only be annotated for patients or their relatives, not for the treating care providers themselves.

- Education, voluntary roles, religions and hobbies also fall under this label, including a mention of a school year. Occupations and job titles of relatives must also be annotated.
- Annotate the full job title, e.g. retired firefighter.
- When an organisation is clearly named, that part falls under Organization.

Examples:

- 1) "The patient works as a plumber."
- 2) "His father is a teacher."
- 3) "She is a hairdresser by profession."
- 4) "The patient is a retired postman." (the full title → 'retired' is part of it)
- 5) "He is in 2nd grade at Regenboog Primary School" (2nd grade = Profession, Regenboog Primary School = Organization:Other)

- 6) "She does **break-dancing** and **art school**" (break-dancing = Profession, art school = Profession)
- 7) "The patient is an **engineer** at **BASF**." (engineer = Profession, BASF = Organization:Other)
- 8) "His brother works as a **chef** at **De Gouden Lepel restaurant**." (chef = Profession, De Gouden Lepel restaurant = Organization:Other)
- 9) "The wife is a **nurse**." (because she is a relative of the patient, not a care provider in the record)
- 10) "The patient worked as a **doctor** in Congo for many years." (the patient's occupation, but not the country)
- 11) "Seen by **dr. Peeters** in the emergency department." (care provider in the record; dr. Peeters = Name:Caregiver)
- 12) "The report was drawn up by the radiologist." (care provider, so not annotated)
- 13) "Discussed with the head of department." (role of a care provider, so not annotated)
- 14) "The patient is the only **imam of the mosque** in **Merksem**." (imam of the mosque = Profession, Merksem = Address_Location:Patient)
- 15) "She is currently studying **communication sciences** at **AP University College**" (communication sciences = Profession, AP University College = Organization:Other)
- 16) "The patient is **self-employed**" (self-employed = Profession)
- 17) "He is **chair of the local pigeon racing club**." (= Profession, because no organisation is clearly named)
- 18) "The patient is the son of **the mayor** of **Ekeren**." (the mayor = Profession, Ekeren = Address_Location:Other)
- 19) "Religion: **Protestantism**" (Protestantism = Profession)

Contactdetails

These are contact details used for communication, excluding postal addresses; postal addresses fall under Address_Location. This includes email, telephone, fax, pagers, URLs, ...

- URLs containing patient-specific IDs fall under ID:Patient

Examples:

- 1) "Telephone number: **03/123.45.67**."
- 2) "Email: **jan.jansen@hotmail.com**."
- 3) "Fax: **03/765.43.21**."
- 4) "Reachable on mobile: **+32 477 12 34 56**."
- 5) "Contact us via **www.ziekenhuis-antwerpen.be**."
- 6) "DECT: **#4521**." (internal telephone number)
- 7) "Practice website: **http://www.huisarts-peeters.be**."
- 8) "The patient lives at **Langestraat 128**." (Langestraat 128 = Address_Location:Patient)
- 9) "Further information can be found at **www.uza.be/cfs-guideline**" (a URL is an indirect contact detail)

ID

This covers all codes and identifiers that can be traced back to one unique person or to a group of people (e.g. a protocol). A distinction is made between ID:Patient and ID:Caregiver. There is currently no reason to define ID:Other as well; as long as that does not change, annotate such cases as Anonymize_Other (see below).

Files without a reference to a care institution or a patient must not be annotated. A reference to a care institution falls under Anonymize_Other.

ID:Patient

All IDs specific to patients, for example patient numbers, national registry numbers, etc.

- Codes for genes or biomarkers do not fall under this label.
- Codes and links for imaging and pathology do fall under this label.
- Codes for surgical materials, diagnoses and systems do not fall under this label if they cannot be traced back to a patient or an organisation.
- All study protocols are also treated as ID:Patient.
- Files that refer to patients also fall under this label.

Examples:

- 1) "Patient number: 12345678."
- 2) "National registry number: 85.07.15-123.45."
- 3) "Record number: UZ-2021-998877."
- 4) "The patient's internal ID is PT-00987."
- 5) "National social security number: 75021512345."
- 6) "Medical imaging can be found at: <https://beelden.zna.be/23h3H0>" (include the full URL, because it identifies the hospital)
- 7) "OR no. H3038L93"
- 8) "Hospitalisation ID: 308354043"
- 9) "Final report 304332-UZA-MDT.pdf" (this refers to the patient)
- 10) "Lab results can be identified by MI 3023 3920"
- 11) "CFDNA 32038" (CFDNA still carries clinical information = cell-free DNA)
- 12) "The patient is taking part in the BHEALTH study under reference K20214" (BHEALTH = ID:Patient, K20214 = ID:Patient)
- 13) "The ICD D633-22042 is implanted" (an ICD device code is a serial number, a unique number traceable to the patient)

ID:Caregiver

All IDs specific to a care provider, for example a national provider registration number. This can be extended to IDs specific to a care institution.

Examples:

- 1) "National provider registration number: 12345678901."
- 2) "Healthcare professional registration number: 19001234501."
- 3) "Doctor ID: DR-998877."
- 4) "The nursing registration number is VRN-223344."

- 5) "The ISO 12452 standard is met, with national accreditation under MED-245" (the ISO number is not annotated, because it cannot be traced back to a care institution)
- 6) "The material has LOT number 23532"

Anonymize Other

This can be seen as the catch-all bin for personally identifiable information about patients or care providers that does not immediately fit another label. Its main purpose is to give annotators the opportunity to flag gaps in the current ProductionLabels. Annotators are always asked to explain in a free-text field why that particular entity was annotated. Always consider whether the piece of information disproportionately increases the chance of re-identification.

Examples:

- 1) "She was admitted after the bus accident in Sierre." (Sierre = Address_Location:Patient; 'bus accident' is generic enough that removing 'Sierre' keeps the re-identification risk from increasing disproportionately)